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1 Introduction

This document provides usage information for Thermal Management feature.

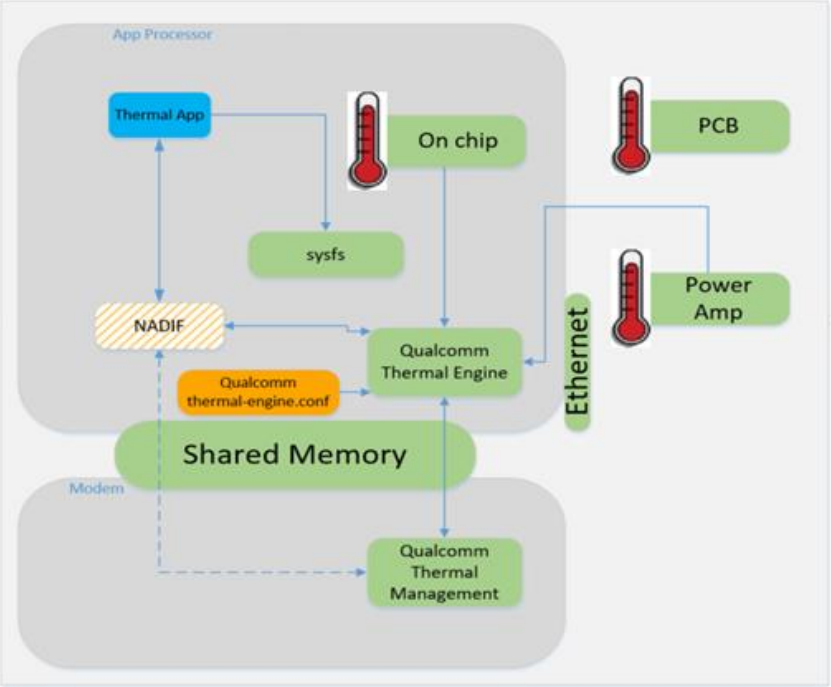

Figure 1 – Thermal Management block diagram

The Thermal Management feature allows a project to control services that may operate when temperature thresholds are exceeded. Telematics Platform Thermal Management provides temperature information from various temperature sensors, modem thermal mitigation status reporting, and application processor wake-up on modem thermal events. The modem thermal mitigation strategy gradually limits/restricts cellular operation to reduce the heat generated by the module (for example cellular carrier aggregation disable, packet data throttling, send SMS disable, and disable all voice calls other than to 112/911) as the temperature of the module rises. In order to support private emergency calls, the most restrictive modem thermal mitigation level (level 3) is disabled. Mitigation level 3 restricts modem operation to emergency calls only, with no SMS or packet data allowed. Private emergency calls, though, may require availability of SMS and/or packet data. NADIF will report when mitigation level 3 temperature threshold is reached. So, the project must monitor NADIF mitigation level reports and take appropriate action to limit activities to support emergency calls only when mitigation level 3 is reported. Additionally, NADIF provides a mitigation level 4 report when modem temperature is approaching critical values to allow application to perform controlled shutdown prior to thermal shutdown. *This feature does not disable thermal mitigation provided by the hardware supplier that protects physical parts through shutdown of the part.*

1.1 Notational Conventions

Ctrl+x	*A sequence with a '+' indicates that you must press and hold one key or pointing device button while pressing another key or pointing device button*
[ ]	*Optional elements in a command*
…..	*Indicates one of the following possibilities:* • *Additional optional arguments in a statement have been omitted* • *The preceding item or items can be repeated one or more times* • *Additional parameters, values, or other information can be entered*

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-	*Line continuation: indicates that a command line or code line continues on the following line when at the end of a line*
Italic text	*Indicates important information, complete titles of manuals or variables*
`Mono-spaced`	*Example code or interactive screen display/output*
Numbers	*All numbers are assumed to be decimal unless otherwise noted. Non decimal radixes – binary, octal or hexadecimal – are explicitly indicated*
Underline blue text	*A link to an external reference (e.g., website, document)*

`Modules`
`|-> section`
`|-> item`
	A link to a section or item in the Telematics Platform html interface specification (Doxygen output from platform code headers)

Diagram Legend

- Telematics Platform
- Product Specific
- 3rd Party

2 Setup

This section describes the testing equipment and hardware needed to develop and test this feature. The setup instructions described here are for reference; and does not imply this is the only setup that will work. The below diagram depicts the test environment with Thermotron (Thermal Chamber).

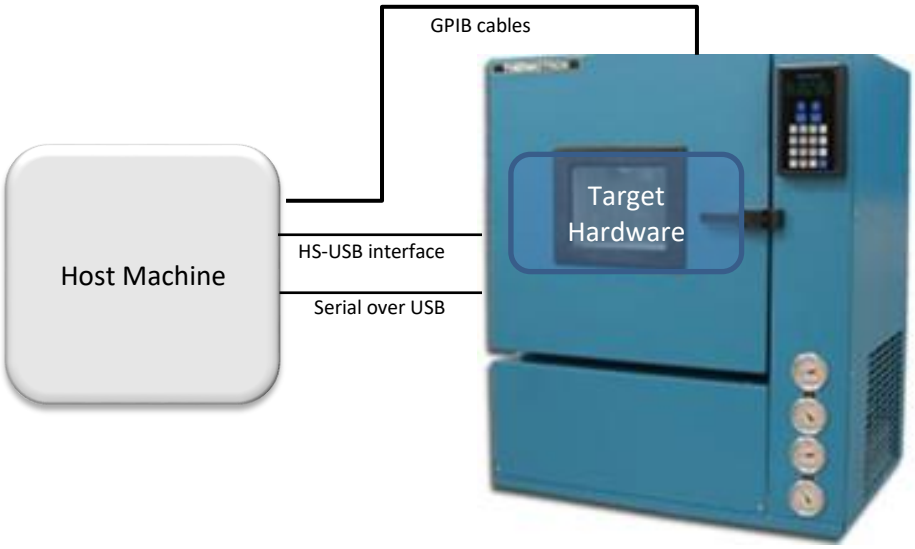

Figure 2 - Test setup with Thermotron

Target hardware is placed inside the Thermotron. Platform uses automated test scripts to control Thermotron functionalities over GPIB interface.

2.1 Tools/Equipment Needed

A thermal chamber is needed to test this feature.

The platform tests with the following equipment:

- Thermal Chamber - Thermotron 2800 (or Thermotron 3200)

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- GPIB cable - GPIB-USB-B

GPIB cables are used to automate Thermotron functionalities.

Platform uses automated scripts to control thermal chamber and validate feature functionality on specified temperature thresholds.

2.2 Setup Instructions

Thermal chamber (Thermotron 2800 / 3200) user manual can be found as below

- https://archive.org/stream/ThermotronIndustries-Thermotron-2800Programmer-Controller.-OperatorManual#page/n9/mode/2up
- http://thermotron.com/pdf/_service/software-updates/3200-manual.pdf

GPIB-USB-B cable and compatible drivers can be found as below (Required if wish to automate thermal chambers functionalities)

- http://www.ni.com/download/ni-visa-17.0/6646/en/
- https://www.ni.com/getting-started/set-up-hardware/instrument-control/gpib-usb

2.3 Setup Troubleshooting

Not Applicable

3 Features/Components

3.1 Reading Thermistor and TSENS Temperature

3.1.1 Overview

This section provides the instruction to read Thermistors and TSENS temperature on Continental 5G NAD module.

3.1.1.1 Thermal Zones for Thermistor Sensor and PMIC Temperature

Continental 5G NAD module has four external thermistors to monitor the temperature. Two thermistors are located on the PCB near the power amplifiers (PA). The third thermistor provides NAD external case temperature. The fourth thermistor provides NAD ambient temperature, which detects the air temperature inside the module.

The temperature of each thermistor is recorded and stored in sysfs in different thermal zones. There are total of 36 thermal zones and temperatures of all thermistors of Continental 5G NAD module locate at zones 35, 36, 37 and 38. Each thermal zone contains two key values: [type] which can be used to determine the thermal type and, [temp] is the recorded temperature, which is read from ADC and the value is in millidegree Celsius. The information of thermistor thermal zones and path to sysfs are provided as below:

Sensor	Units	Sysfs path
Power Amp #1	Millidegree Celsius	/sys/class/thermal/thermal_zone0 (see note 1) /sys/class/thermal/thermal_zone35 (see note 2)
Power Amp #2	Millidegree Celsius	/sys/class/thermal/thermal_zone1 (see note 1) /sys/class/thermal/thermal_zone36 (see note2)
NAD case	Millidegree Celsius	/sys/class/thermal/thermal_zone37
NAD ambient	Millidegree Celsius	/sys/class/thermal/thermal_zone38

Note 1: these thermal zones provide instantaneous temperature

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Note 2: these thermal zones provide polling temperature at “0” polling-delay interval

Sensor	Units	Sysfs path
PMIC	Millidegree Celsius	/sys/class/thermal/thermal_zone33

Note: the above thermal zone information of all thermistors can be used when using releases conmod-sa515m-3.3.302.0 or later. For releases prior conmod-sa515m-3.3.302.0, the thermal zone information is used as indicated in the table below. The thermal zone for PMIC temperature information can’t be used when using releases earlier than conmod-sa515m-3.3.302.0.

Sensor	Units	Sysfs path
Power Amp #1	Millidegree Celsius	/sys/class/thermal/thermal_zone42
Power Amp #2	Millidegree Celsius	/sys/class/thermal/thermal_zone43
NAD case	Millidegree Celsius	/sys/class/thermal/thermal_zone44
NAD ambient	Millidegree Celsius	/sys/class/thermal/thermal_zone45

3.1.1.2 Thermal Zones for TSENS Sensor Temperature

Continental 5G NAD module has seven TSENS sensors to monitor the temperature. These sensors are On-Die Temperature Sensors located on the modem chipset. These TSENS sensors are as follows:

- MDM CORE.0, MDM_CORE_1 and MDM_CORE 2 are located on Modem Core processor
- MDM_Q6 is located on DSP6 (Audio Processor)
- CPU0.0 is located on the A7 application processor
- IPA is located on the IP accelerator
- AOSS is Always On Sub-System

The temperature of each TSENS is recorded and stored also in sysfs in different thermal zones. There are total of 39 thermal zones and the temperature of all seven TSENS of Continental 5G NAD module are located at zones 16-22. Each thermal zone contains two key values: [type] which can be used to determine the thermal type and, [temp] is the recorded temperature in millidegree Celsius.

Sensor	Units	Sysfs path
Always on Subsystem	Millidegree Celsius	/sys/class/thermal/thermal_zone26
Audio Processor	Millidegree Celsius	/sys/class/thermal/thermal_zone27
IP Accelerator	Millidegree Celsius	/sys/class/thermal/thermal_zone28
A7 Application Processor	Millidegree Celsius	/sys/class/thermal/thermal_zone29
Modem Core #2	Millidegree Celsius	/sys/class/thermal/thermal_zone30
Modem Core #1	Millidegree Celsius	/sys/class/thermal/thermal_zone31
Modem Core #0	Millidegree Celsius	/sys/class/thermal/thermal_zone32

3.1.1.3 Product Expectations

Not Applicable

3.1.1.4 Extending Platform Functionality

Not Applicable

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3.1.2 Setup

3.1.2.1 Tools/Equipment Needed

Not Applicable

3.1.2.2 Setup Instructions

Not Applicable

3.1.2.3 Setup Troubleshooting

Not Applicable

3.1.3 Usage

3.1.3.1 Basic Instructions

Not Applicable

3.1.3.2 Detailed Instructions

3.1.3.2.1 How to obtain Thermistor Sensor temperature from sysfs file when using any release above or equal conmod-sa515m-3.3.302.0 release.

The recorded temperature of each thermistor is stored in sysfs. Instructions to read the temperature from an ADB shell window is provided below:

1. go to adb shell window
2. from adb shell window, go to thermal-zone folder of sysfs. See the screenshot below:

```
[root@sa515m ~]# cd /sys/class/thermal/
[root@sa515m thermal]# ls
cooling_device0   cooling_device6   thermal_zone19    thermal_zone32
cooling_device1   cooling_device7   thermal_zone2     thermal_zone33
cooling_device10  cooling_device8   thermal_zone20    thermal_zone34
cooling_device11  cooling_device9   thermal_zone21    thermal_zone35
cooling_device12  thermal_zone0     thermal_zone22    thermal_zone36
cooling_device13  thermal_zone1     thermal_zone23    thermal_zone37
cooling_device14  thermal_zone10    thermal_zone24    thermal_zone38
cooling_device15  thermal_zone11    thermal_zone25    thermal_zone4
cooling_device16  thermal_zone12    thermal_zone26    thermal_zone5
cooling_device17  thermal_zone13    thermal_zone27    thermal_zone6
cooling_device18  thermal_zone14    thermal_zone28    thermal_zone7
cooling_device2   thermal_zone15    thermal_zone29    thermal_zone8
cooling_device3   thermal_zone16    thermal_zone3     thermal_zone9
cooling_device4   thermal_zone17    thermal_zone30
cooling_device5   thermal_zone18    thermal_zone31
```


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3. Change path to thermal_zone35 to access pa-therm1-usr thermistor sensor temperature (PA_THERM1)

```
[root@sa515m thermal]# cd thermal_zone35/
[root@sa515m thermal_zone35]# ls
available_policies  offset              subsystem
integral_cutoff     passive             sustainable_power
k_d                 passive_delay       temp
k_i                 policy              type
k_po                polling_delay       uevent
k_pu                power
mode                slope
[root@sa515m thermal_zone35]# cat type
pa-therm1-usr
[root@sa515m thermal_zone35]# cat temp
26724
[root@sa515m thermal_zone35]#
```

4. Change path to thermal_zone36 to access pa-therm2-usr thermistor sensor temperature (PA_THERM2)

```
[root@sa515m thermal]# cd thermal_zone36/
[root@sa515m thermal_zone36]# ls
available_policies  offset              subsystem
integral_cutoff     passive             sustainable_power
k_d                 passive_delay       temp
k_i                 policy              type
k_po                polling_delay       uevent
k_pu                power
mode                slope
[root@sa515m thermal_zone36]# cat type
pa-therm2-usr
[root@sa515m thermal_zone36]# cat temp
27280
[root@sa515m thermal_zone36]#
```

5. Change path to thermal_zone37 to access sdx-case-them-usr thermistor sensor temperature

```
[root@sa515m thermal]# cd thermal_zone37/
[root@sa515m thermal_zone37]# ls
available_policies  offset              subsystem
integral_cutoff     passive             sustainable_power
k_d                 passive_delay       temp
k_i                 policy              type
k_po                polling_delay       uevent
k_pu                power
mode                slope
[root@sa515m thermal_zone37]# cat type
sdx-case-therm-usr
[root@sa515m thermal_zone37]# cat temp
28051
[root@sa515m thermal_zone37]#
```


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6. Change path to thermal_zone38 to access ambient-them-usr thermistor sensor temperature

```
[root@sa515m thermal]# cd thermal_zone38/
[root@sa515m thermal_zone38]# ls
available_policies  offset              subsystem
integral_cutoff     passive             sustainable_power
k_d                 passive_delay       temp
k_i                 policy              type
k_po                polling_delay       uevent
k_pu                power
mode                slope
[root@sa515m thermal_zone38]# cat type
ambient-therm-usr
[root@sa515m thermal_zone38]# cat temp
28301
[root@sa515m thermal_zone38]#
```

Note: sysfs nodes can be accessed directly from application software.

3.1.3.2.2 How to obtain TSENS Sensor temperature from sysfs File

The recorded temperature of each TSENS is stored in sysfs.

Instructions to read the temperature from an ADB shell window is provided below:

1. go to adb shell window
2. from adb shell window, go to thermal-zone folder of sysfs. See the screenshot below:

```
[root@sa515m ]# cd /sys/class/thermal/
[root@sa515m thermal]# ls
cooling_device0   cooling_device6   thermal_zone19    thermal_zone32
cooling_device1   cooling_device7   thermal_zone2     thermal_zone33
cooling_device10  cooling_device8   thermal_zone20    thermal_zone34
cooling_device11  cooling_device9   thermal_zone21    thermal_zone35
cooling_device12  thermal_zone0     thermal_zone22    thermal_zone36
cooling_device13  thermal_zone1     thermal_zone23    thermal_zone37
cooling_device14  thermal_zone10    thermal_zone24    thermal_zone38
cooling_device15  thermal_zone11    thermal_zone25    thermal_zone4
cooling_device16  thermal_zone12    thermal_zone26    thermal_zone5
cooling_device17  thermal_zone13    thermal_zone27    thermal_zone6
cooling_device18  thermal_zone14    thermal_zone28    thermal_zone7
cooling_device2   thermal_zone15    thermal_zone29    thermal_zone8
cooling_device3   thermal_zone16    thermal_zone3     thermal_zone9
cooling_device4   thermal_zone17    thermal_zone30
cooling_device5   thermal_zone18    thermal_zone31
```


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3. Change path to thermal_zone26 to access aoss0-usr TSENS sensor temperature

```
[root@sa515m thermal]# cd thermal_zone26/
[root@sa515m thermal_zone26]# ls
available_policies  passive_delay       trip_point_0_hyst
integral_cutoff     policy              trip_point_0_temp
k_d                 polling_delay       trip_point_0_type
k_i                 power               trip_point_1_hyst
k_po                slope               trip_point_1_temp
k_pu                subsystem           trip_point_1_type
mode                sustainable_power   type
offset              temp                uevent
[root@sa515m thermal_zone26]# cat type
aoss0-usr
[root@sa515m thermal_zone26]# cat temp
29900
```

4. Change path to thermal_zone27 to access mdm-q6-usr TSENS sensor temperature

```
[root@sa515m thermal]# cd thermal_zone27/
[root@sa515m thermal_zone27]# ls
available_policies  passive_delay       trip_point_0_hyst
integral_cutoff     policy              trip_point_0_temp
k_d                 polling_delay       trip_point_0_type
k_i                 power               trip_point_1_hyst
k_po                slope               trip_point_1_temp
k_pu                subsystem           trip_point_1_type
mode                sustainable_power   type
offset              temp                uevent
[root@sa515m thermal_zone27]# cat type
mdm-q6-usr
[root@sa515m thermal_zone27]# cat temp
30300
[root@sa515m thermal_zone27]#
```

5. Change path to thermal_zone28 to access ipa-usr TSENS sensor temperature

```
[root@sa515m thermal]# cd thermal_zone28/
[root@sa515m thermal_zone28]# ls
available_policies  passive_delay       trip_point_0_hyst
integral_cutoff     policy              trip_point_0_temp
k_d                 polling_delay       trip_point_0_type
k_i                 power               trip_point_1_hyst
k_po                slope               trip_point_1_temp
k_pu                subsystem           trip_point_1_type
mode                sustainable_power   type
offset              temp                uevent
[root@sa515m thermal_zone28]# cat type
ipa-usr
[root@sa515m thermal_zone28]# cat temp
31100
[root@sa515m thermal_zone28]#
```


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6. Change path to thermal_zone29 to access cpu0-0-usr TSENS sensor temperature

```
[root@sa515m thermal]# cd thermal_zone29/
[root@sa515m thermal_zone29]# ls
available_policies  passive_delay       trip_point_0_hyst
integral_cutoff     policy              trip_point_0_temp
k_d                 polling_delay       trip_point_0_type
k_i                 power               trip_point_1_hyst
k_po                slope               trip_point_1_temp
k_pu                subsystem           trip_point_1_type
mode                sustainable_power   type
offset              temp                uevent
[root@sa515m thermal_zone29]# cat type
cpu0-0-usr
[root@sa515m thermal_zone29]# cat temp
31100
[root@sa515m thermal_zone29]#
```

7. Change path to thermal_zone30 to mdm-core-2-usr TSENS sensor temperature

```
[root@sa515m thermal]# cd thermal_zone30/
[root@sa515m thermal_zone30]# ls
available_policies  passive_delay       trip_point_0_hyst
integral_cutoff     policy              trip_point_0_temp
k_d                 polling_delay       trip_point_0_type
k_i                 power               trip_point_1_hyst
k_po                slope               trip_point_1_temp
k_pu                subsystem           trip_point_1_type
mode                sustainable_power   type
offset              temp                uevent
[root@sa515m thermal_zone30]# cat type
mdm-core-2-usr
[root@sa515m thermal_zone30]# cat temp
30700
[root@sa515m thermal_zone30]#
```

8. Change path to thermal-zone31 to access mdm-core-1-usr TSENS sensor temperature

```
[root@sa515m thermal]# cd thermal_zone31/
[root@sa515m thermal_zone31]# ls
available_policies  passive_delay       trip_point_0_hyst
integral_cutoff     policy              trip_point_0_temp
k_d                 polling_delay       trip_point_0_type
k_i                 power               trip_point_1_hyst
k_po                slope               trip_point_1_temp
k_pu                subsystem           trip_point_1_type
mode                sustainable_power   type
offset              temp                uevent
[root@sa515m thermal_zone31]# cat type
mdm-core-1-usr
[root@sa515m thermal_zone31]# cat temp
31500
[root@sa515m thermal_zone31]#
```


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9. Change path to thermal-zone32 to access mdm-core-0-usr TSENS sensor temperature

```
[root@sa515m thermal]# cd thermal_zone32
[root@sa515m thermal_zone32]# ls
available_policies  passive_delay       trip_point_0_hyst
integral_cutoff     policy              trip_point_0_temp
k_d                 polling_delay       trip_point_0_type
k_i                 power               trip_point_1_hyst
k_po                slope               trip_point_1_temp
k_pu                subsystem           trip_point_1_type
mode                sustainable_power   type
offset              temp                uevent
[root@sa515m thermal_zone32]# cat type
mdm-core-0-usr
[root@sa515m thermal_zone32]# cat temp
30700
[root@sa515m thermal_zone32]#
```

Notes:

- sysfs nodes can be accessed directly from application software
- Follow the same steps as provided in the above for the instructions to read thermistor temperatures when using conmod-sa515m-3.3.302.0 or later releases. The only difference is the thermal zone node which is slightly different. Additional instructions are provided as below to read temperature for Power Amp #1 from thermal_zone0 and thermal_zone35, and Power Amp #2 from thermal_zone1 and thermal_zone36.

```
[root@sa515m ]# cd /sys/class/thermal/thermal_zone0/
[root@sa515m thermal_zone0]# cat
available_policies  mode                slope               trip_point_0_type
integral_cutoff     offset              subsystem/          type
k_d                 passive_delay       sustainable_power   uevent
k_i                 policy              temp
k_po                polling_delay       trip_point_0_hyst
k_pu                power/              trip_point_0_temp
[root@sa515m thermal_zone0]# cat type
modem-lte-sub6-pa1
[root@sa515m thermal_zone0]# cd /
[root@sa515m ]# cd /sys/class/thermal/thermal_zone0/
[root@sa515m thermal_zone0]# cat type
modem-lte-sub6-pa1
[root@sa515m thermal_zone0]# cat temp
26000
[root@sa515m thermal_zone0]# cd ../thermal_zone35/
[root@sa515m thermal_zone35]# cat type
pa-therm1-usr
[root@sa515m thermal_zone35]# cat temp
26613
[root@sa515m thermal_zone35]#
```


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```
[root@sa515m ]# cd /sys/class/thermal/thermal_zone0/
[root@sa515m thermal_zone0]# cat type
modem-lte-sub6-pa1
[root@sa515m thermal_zone0]# cat temp
26000
[root@sa515m thermal_zone0]# cd ../thermal_zone35/
[root@sa515m thermal_zone35]# cat type
pa-therm1-usr
[root@sa515m thermal_zone35]# cat temp
26613
[root@sa515m thermal_zone35]# cd /
[root@sa515m ]# cd /sys/class/thermal/thermal_zone1/
[root@sa515m thermal_zone1]# cat type
modem-lte-sub6-pa2
[root@sa515m thermal_zone1]# cat temp
26000
[root@sa515m thermal_zone1]# cd ../thermal_zone36/
[root@sa515m thermal_zone36]# cat type
pa-therm2-usr
[root@sa515m thermal_zone36]# cat temp
27078
[root@sa515m thermal_zone36]#
```

- Below provides the instructions to read PMIC temperature.

```
[root@sa515m ]# cd /sys/class/thermal/thermal_zone33/
[root@sa515m thermal_zone33]# cat type
pmxprairie_tz
[root@sa515m thermal_zone33]# cat temp
31261
[root@sa515m thermal_zone33]#
```

3.1.3.3 Testing/Error Reporting

Not Applicable

3.2 Modem Thermal Mitigation State Notifications

3.2.1 Overview

3.2.1.1 Platform Content Overview

Platform provides application access to modem thermal mitigation status for the PA and Modem junction temperatures. The status of each mitigation device can be provided synchronously when queried/requested via a NADIF API as well as asynchronously when thermal mitigation level changes via a NADIF notification when enabled.

Continental 5G NAD module supports three distinct thermal mitigation devices and all the mitigation operations are performed in the modem processor. Below describes the functionality of each mitigation device.

- PA mitigation device is to support 2G/3G and 4G for the uplink packet data throttling and Power back-off
- PA_FR1 mitigation device is to support 5G NR/Sub6 for the uplink packet data throttling and Power back-off
- Modem mitigation device is to support the downlink packet data throttling for all RATs

3.2.1.1.1 Uplink Mitigation

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PA and PA_FR1 mitigation devices use common PA mitigation level 1 mitigation action but mitigation level 2 action is slightly different. Modem performs LTE fallback for the uplink when handling PA_FR1 mitigation.

Level	PA Mitigation Actions	PA_FR1 Mitigation Actions
0	• No Mitigation	• No Mitigation
1	• Uplink packet data throttling	• Uplink packet data throttling
2	• Continue uplink packet data throttling • Enable Power back-off • Limit maximum Tx power (23 dBm)	• Power back off duty cycle • LTE fallback

The action of PA and PA_FR1 Mitigation levels 1 and 2 are performed by the Modem according to the threshold settings.

3.2.1.1.2 Downlink Mitigation

Modem mitigation device supports mitigation levels and actions as follows:

Level	LTE Mitigation Actions	5G NR/Sub6 Modem Mitigation Actions
0	• No Mitigation	• No Mitigation
1	• Not Applicable	• Downlink packet data throttling • Perform 5G Non Standalone (NSA) /Sub6 fallback to LTE • Disable 5G Standalone (SA)/Sub6
2	• Downlink data throttling • Perform 4x to 2x MIMO fallback • Limit Carrier Aggregation (CA) as needed.	• Downlink packet data throttling • Perform 4x to 2x MIMO fallback • Drop to LTE 1 Carrier Component 2x MIMO

The action of Modem Mitigation levels 1 and 2 are performed by the modem according to the threshold settings. Mitigation level 1 for LTE is not applicable. Note that proper mitigation action will be performed by the modem with respect to which IRAT the modem is operating on.

3.2.1.2 Product Expectations

Application should monitor Mitigation level and take an appropriate action when temperature thresholds are exceeded.

Modem mitigation level 3 is disabled, to support private emergency calls (e.g., emergency calls to a Telematics Service Center), which may require SMS and/or packet data support. The Application is responsible for monitoring temperature levels and terminating all non-emergency services (e.g., Wi-Fi, voice calls, packet data connections, etc. not related to an emergency call) when mitigation level 3 temperature threshold is reached. Monitoring of temperature levels can be done via thermistor and TSENS temperature (see section 3.1.1). Non-emergency services should not be restarted until the temperature drops below the mitigation level 3 hysteresis threshold.

Notes:

NADIF will report modem mitigation levels 3, 4 for thermistors and TSENSE devices (see section 3.1.1)..

3.2.1.3 Extending Platform Functionality

Not Applicable

3.2.2 Setup

3.2.2.1 Tools/Equipment Needed

Not applicable

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3.2.2.2 Setup Instructions

Not applicable

3.2.2.3 Setup Troubleshooting

Not Applicable

3.2.3 Usage

3.2.3.1 Basic Instructions

Not Applicable

3.2.3.2 Detailed Instructions

For more details on these APIs and types see Doxygen links in the following table:

API/Type Description	Doxygen Path
Structure of notification request from Modem	Modules |-> Modem |-> Nadif |-> Modem Status |-> **NOTIFICATION_REQUEST_DATA**
API to register / deregister for modem thermal notifications	Modules |-> Modem |-> Nadif |-> Modem Status |-> **NADIF_modem_set_thermal_mitigation_report()**
API to query current modem thermal mitigation level	Modules |-> Modem |-> Nadif |-> Modem Status |-> **NADIF_modem_get_thermal_mitigation_level()**
API to query supported mitigation device list	Modules |-> Modem |-> Nadif |-> Modem Status |-> **NADIF_modem_get_thermal_mitigation_device_list()**
Notification data type of API	Modules |-> Modem |-> Nadif |-> NADIF Common Types |-> **NADIF_FEAT_STATUS_MODEM_THERMAL_MITIGATION_CHANGE**

<u>Mitigation Device List</u>

The API `NADIF_modem_get_thermal_mitigation_device_list()` is used to request a list of all thermal mitigation devices supported by the modem. The API returns a list of mitigation devices by name (e.g., character string) and the maximum mitigation level supported/reported by each device. Note that this is a synchronous API.

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Continental 5G NAD module supports three mitigation devices and they are specified by device name as follows:

Mitigation Device Name	Mitigation Functionality
pa	Support 2G/3G/4G uplink data throttling and tx power backoff
pa_fr1	Support 5G/Sub6 uplink data throttling and tx power backoff
modem	Support downlink data throttling for all RATs

Mitigation Level Query

The API NADIF_modem_get_thermal_mitigation_level() is used to request the current modem thermal mitigation level for a mitigation device, thermistor, TSENS. The device name is legacy parameter that is not effectively used but has to be provided in the API and must match one of the device names in the mitigation device list returned by NADIF_modem_get_thermal_mitigation_device_list(). Response will be received with NADIF_FEAT_MODEM_GET_THERMAL_MITIGATION_LEVEL feature ID. Mitigation Level Change Notification

The API `NADIF_modem_set_thermal_mitigation_report()` is used to register or deregister for asynchronous modem thermal notifications. When modem thermal notification report is registered, mitigation state change for each mitigation device will be notified to the registered application. The modem thermal notification message will contain a `NADIF_FEAT_STATUS_MODEM_THERMAL_MITIGATION_CHANGE` feature ID. The application can process this message to obtain modem thermal mitigation level change information.

Note: `NADIF_modem_set_thermal_mitigation_report()` will return a success indication when invoked to register for modem thermal notifications if at least one mitigation device is available from the modem. In other words, if one device is unavailable while other devices are available the API will return a success indication.

Special consideration for modules using Continental modem based BSP/HAL

When the NADIF thermal status change notification is enabled, this status will be provided regardless of application state. So, a modem thermal mitigation level change would cause the application to wake-up if application is asleep/suspended. If there is no need to track modem thermal mitigation level status changes while in a low power mode, then modem thermal mitigation level change notifications should be disabled prior to mode entry (e.g., before going to sleep).

3.2.3.3 Note: This is not applied to mitigation levels 3,4.Testing/Error Reporting

1. From lte-test, use option 104 to enable modem thermal mitigation event report.

```
[root@sa515m ]#sudo -u modem lte-test -i
sudo -u modem lte-test -i
0) Exit
1) Voice Dial
2) Voice Hangup
3) Voice Answer
4) Get Voice call status
5) Create/Modify profile
6) Get profile
7) Data start
8) Data stop
9) Data Get status
10) SMS send
11) SMS read
```


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```
103) Get modem thermal mitigation level
104) Set modem thermal mitigation event report
105) Send APDU command to the SIM/UICC
106) Get On-device Logging parameters
107) Inband modem
108) Set data for Inband Modem
109) Set eCall only emulation
110) Get eCall only emulation status
111) Answer incoming inband call
112) voice dial over IMS Custom user data using SIP
113) Send RAW apdu data over logical channel
114) Open logical channel to send APDU data
115) Close APDU logical channel
116) Set Remote profile switch lock configuration
117) Get Remote profile switch lock configuration
118) Client release
119) Client release multiple
120) Client init multiple
121) Get server socket data
122) Read/Write NV items
123) Send Diag command to Modem
124) Get All Voice call status
125) SMS read PDU
126) Set physical sim slot
127) Voice manage calls
128) Set persistency Key
129) Get persistency Key
130) Set Broadcast Messages
131) Set MIPI registers
132) Get PDN throttle info
133) Stop MSD transmission
134) Get Broadcast Messages
135) Get IMS configuration
136) Get emergency phone numbers list
137) Set Manual or Automatci MSD transmission Setting
138) Set Domain Preferences
139) Get Domain Preferences
104
104
Select an option:
1) Enable
2) Disable
```

2. Heat up the module to bring temperature to threshold level 1, level 2.
3. Verify NADIF can obtain notification of correct Mitigation level change from Modem.

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3.3 Modem Thermal Wakeup

3.3.1 Overview

3.3.1.1 Platform Content Overview

As part of this feature, platform adds capability to resume application processor from low power mode when temperature thresholds are reached. The temperature thresholds are defined in persistency data for PA thermistors and Modem TSENs. They represent two values high threshold and low threshold, and each will be used by the modem to determine when the temperature of PA thermistors and Modem TSENs to trigger wakeup of the application processor.

Below describes the threshold values defined for PA thermistors and Modem TSENs in Persistency data.

Sensor	Parameter from Persistency	Threshold Value
Power Amp #1	pa-therm-high-threshold	90 degree C
	pa-therm-low-threshold	-40 degree C
Power Amp #2	pa_1_therm-high-threshold	90 degree C
	pa_1-therm-low-threshold	-40 degree C
Modem Core 0	modem_tsens-therm-high-threshold	95 degree C
	modem_tsens-therm-low-threshold	-40 degree C
Modem Core 1	modem_tsens1-therm-high-threshold	95 degree C
	Modem_tsens1-therm-low-threshold	-40 degree C

Note: Modem core 2 high/low thresholds are not provided. The modem does not support monitoring of this sensor.

3.3.1.2 Product Expectations

Temperature thresholds are typically configured in the persistency and will be sent to the modem during power up initialization. Persistency data is run-time parameter, but value changes will not be reflected in the modem until power cycle. Changing of these values during run-time is not recommended.

3.3.1.3 Extending Platform Functionality

Not Applicable

3.3.2 Setup

3.3.2.1 Tools/Equipment Needed

Not Applicable

3.3.2.2 Setup Instruction

Not Applicable

3.3.2.3 Setup Troubleshooting

Not Applicable

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3.3.3 Usage

3.3.3.1 Basic Instructions

Not Applicable

3.3.3.2 Detailed Instructions

The NADIF wakeup reason `NADIF_WU_THERMAL_THRESHOLD` should be enabled prior to entering module low power mode. When a modem thermal event occurs (e.g., modem mitigation level change), the application processor will be awakened and NADIF will report the wakeup reason as `NADIF_WU_THERMAL_THRESHOLD` when wakeup reason is queried.

Refer to "Low Power Mode – suspend/resume" section in TP_F101U-00039 for details on low power mode operation, including enabling NADIF wakeup reasons prior to mode entry.

The application processor can query API NADIF_modem_get_thermal_mitigation_level() to obtain the thermal wakeup information after NADIF_WU_THERMAL_THRESHOLD is reported.

3.3.3.3 Testing/Error Reporting

Test Instructions:

1. Set thermal wake up reason `NADIF_WU_THERMAL_THRESHOLD` prior to entering low power mode as specified in "Low Power Mode – suspend/resume" section in TP_F101U-00039.
2. Enter low power mode (e.g., suspend application processor) as specified in "Low Power Mode – suspend/resume" section in TP_F101U-00039.
3. Confirm application processor is suspended.
4. Use thermal chamber to heat up the device and bring temperature to reach the threshold for modem mitigation level change.
5. Observe application processor wakes up.
6. Query NADIF wakeup reason and confirm `NADIF_WU_THERMAL_THRESHOLD` is reported as specified in "Low Power Mode – suspend/resume" section in TP_F101U-00039.

3.4 Non-emergency calls verification with Modem Thermal Mitigation Stage 3 disabled

3.4.1 Overview

3.4.1.1 Platform Content Overview

With PA Thermal mitigation Stage 3 disabled, voice call, eCall, SMS, packet data will be supported when NAD successfully registers with the Network.

3.4.1.2 Product Expectations

Application should monitor Mitigation level and take appropriate action when temperature thresholds are exceeded.

Modem mitigation level 3 is disabled, to support private emergency calls (e.g., emergency calls to a Telematics Service Center), which may require SMS and/or packet data support. The Application is responsible for monitoring temperature levels and terminating all non-emergency services (e.g., Wi-Fi, voice calls, packet data connections, etc. not related to an emergency call) when mitigation level 3 temperature threshold is reached. Monitoring of temperature levels can be done via thermistor and TSENS temperature (see section 3.1.1). Non-emergency services should not be restarted until the temperature drops below the mitigation level 3 hysteresis threshold.

Note: NADIF will report modem mitigation levels 3, 4 for thermistors and TSENSE devices (see section 3.1.1).

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Verification of non-emergency calls such as Voice, SMS and Data Calls can be performed with a live Network when temperature is nominal, high or critical.

3.4.1.3 Extending Platform Functionality

Not Applicable

3.4.2 Setup

3.4.2.1 Tools/Equipment Needed

Not Applicable

3.4.2.2 Setup Instructions

Not Applicable

3.4.2.3 Setup Troubleshooting

Not Applicable

3.4.3 Usage

3.4.3.1 Basic Instructions

Not applicable

3.4.3.2 Detailed Instructions

1) Verify Voice Call can be originated when temperature below 60, above 80 and above 100
2) Verify Voice call can be terminated when temperature below 60, above 80 and above 100
3) Verify SMS can be sent from the device when temperature below 60, above 80 and above 100
   a. systemctl start lte-test
   b. lte-test -c 10 TEST 6306504629 0 1 2 1 17-10-05-10:17:17
4) Verify SMS can be received by the device when temperature below 60, above 80 and above 100
   a. From a cell phone, send a text message to the device while temperature below 60, above 80 and above 100.
   b. Use lte-test option 11 SMS read and pass SMS index printed in the log for incoming SMS. Check NADIF_STATUS_SMS response
5) Verify packet data can be connected to LTE Network and ping when temperature reaches temperature below 60, above 80 and above 100.

Some example complex test scenarios suggested below for PA Mitigation device require verification using a call box to simulate Network behavior and a thermal chamber to control the temperature. Note that similar test scenarios could be also performed with Modem Mitigation device.

Test Number	Test Scenario	Test Procedures	Expected Result
1	Verify the device does not release the voice and data calls when temperature is	This test can be verified with the Live Network and the device needs to be placed in	With different Mitigation Level change especially when the temperature is above Mitigation State 2

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Test Number	Test Scenario	Test Procedures	Expected Result
	above the Mitigation Level 2 temperature threshold.	the thermal Chamber to control the temperature.	level, the device should not release voice or data calls when call is active.
2	Perform eCall when temperature at state 0, state 1 and state 2 and above state 2.	This test needs to verify on the simulator that supports eCall. The device should be placed in the thermal chamber to control the temperature.	eCall can be placed under any temperature. Transmission of MSD should be completed successfully under any temperature.
3	Perform SRVCC when temperature at state 0, state 1 and state 2 and above state 2.	This test needs to verify on the simulator that supports SRVCC. The device should be placed in the thermal chamber to control the temperature.	VoLTE Call can be established with good audio on 4G and can be successfully switched to 3G, under temperature change to different mitigation state.
4	Perform CSFB (Voice Call handover from 3G to 4G) when temperature at state 0, state 1 and state 2 and above state 2	This test needs to verify on the simulator that supports CFSB. The device should be placed in the thermal chamber to control the temperature.	3G call can be established with good audio on 3G and can be successfully switched to 4G. , under temperature change to different mitigation state.
5	Perform 112/911 emergency call when temperature above Mitigation State 2 and when the device registers with the Network.	This test needs to verify on the simulator that supports emergency 112/911 calls and when the device registers with the Network. The device should be placed in the thermal chamber to control the temperature.	Emergency 112/911 can be placed under temperature change to different mitigation state.
6	Perform 112/911 emergency call when temperature above Mitigation State 2 and when the device in limited service	This test needs to verify on the simulator that supports emergency 112/911 calls and when the device in limited service. Use attenuator to weaken the signal strength forcing the device to be in limited service. The device should be placed in the thermal chamber to control the temperature	Emergency 112/911 can be placed under temperature change to different mitigation state and gets to limited service
7	Perform LTE Attach to the Network when temperature change from Mitigation Level 1 to Mitigation Level2 and Network reject with Attached reject cause 15 (No suitable Cell). Then	This test needs to be verified with the simulator where it can trigger attach reject cause 15. Verify LTE attach becomes successful after removing the trigger of reject cause 15. The device should be placed in the	With different Mitigation Level change, the device can perform LTE registration attempts after being rejected by the

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Test Number	Test Scenario	Test Procedures	Expected Result
	remove the trigger to verify LTE Attach to the Network can be performed successfully	thermal chamber to control the temperature	Network with reject cause 15.
8	Verify the device can be recovered from Service Loss during Mitigation state change from Level 1 to Level 2 and vice versa.	This test needs to be verified with the simulator where signal strength can be reduced to force the device to service loss. The device should be placed in the thermal chamber to control the temperature	With different Mitigation Level change, the device should be able to recover from service loss
9	Verify the device can switch to different LTE Band (B2 to B5) when Mitigation state change from Level 1 to Level 2 and vice versa.	This test needs to be verified on the simulator where the Network can switch to different LTE Band. The device should be placed in the thermal chamber to control the temperature	With different Mitigation Level change, the device should be able to acquire the Network on different LTE band
10	Verify the device operating mode can be switch from airplane mode to online mode and vice versa when Mitigation state change from Level 1 to Level 2 and vice versa.	This test can be verified with the Live Network and the device needs to be placed in the thermal Chamber to control the temperature.	With different Mitigation Level change, the device should be able to switch from airplane mode to online mode.

3.5 Modem Thermal Levels 3 and 4 Reporting

3.5.1 Overview

3.5.1.1 Platform Content Overview

Platform provides application access to modem thermal levels 3 and 4 statuses for all supported devices. These statuses are reports shared by NADIF with no additional mitigation performed by the modem.

The report of modem thermal levels 3 and 4 status of each supported mitigation device can be provided synchronously when queried/requested via a NADIF API.

Modem thermal device supports level 3 and 4 as follows:

Level	Notification Implies	Application action
3	The critical temperature level is reached.	All application services not related to servicing an emergency should be turned off.
4	The extreme temperature level reached.	Modem shutdown imminent; application should prepare for shutdown, such as saving data to non-volatile memory and controlled service shutdown.

The threshold for level 3 and 4 reporting can be set using the persistency.

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Below table describe the threshold values defined for PA thermistors 3 and Modem TSENs level 3 and 4 in Persistency data:

Sensor/Mitigation level	Parameter from Persistency	Threshold Value
Power Amp/Level 3	PA_THERMAL_LEVEL_3	100 degree C
Modem Core/Level 3	TSENS_THERMAL_LEVEL_3	105 degree C
Modem Core/Level 4	TSENS_THERMAL_LEVEL_4	117 degree C

3.5.1.2 Product Expectations

The levels 3 and 4 are for reports only and application is expected to monitor the modem thermal levels 3 and 4 and take appropriate actions when temperature thresholds are exceeded.

NADIF will report modem thermal level 3, when the temperature is above 100 degrees C for the PA thermistors.

NADIF will report modem thermal level 3 when the temperature is above 105 degrees C and level 4 when the temperature is above 117 degrees C for Modem TSENS.

When the application processor is in suspend state, NADIF will not report levels 3 & 4 to the application. NADIF will resume Levels 3 & 4 reporting action once the application is woken up when temperature reaches the wakeup thresholds defined in persistency.

3.5.1.3 Extending Platform Functionality

Not Applicable

3.5.2 Setup

3.5.2.1 Tools/Equipment Needed

Not Applicable

3.5.2.2 Setup Instructions

Not Applicable

3.5.2.3 Setup Troubleshooting

Not Applicable

3.5.3 Usage

3.5.3.1 Basic Instructions

Not applicable

3.5.3.2 Detailed Instructions

See Detailed Instructions for Modem Thermal Mitigation State Notifications.

4 Known Issues

Not Applicable

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5 FAQs

Not Applicable

6 Troubleshooting/Tips and Tricks

Not Applicable

7 References

7.1 Abbreviations/Acronyms/Definitions

Term	Meaning
CSFB	Circuit Switched Fall Back
MDM	Modem Device Management
NAD	Network Access Device
PA	Power Amplifier
SRVCC	Single Radio Voice Call Continuity

7.2 Platform Documentation

TP_102G-00001: Telematics Platform Glossary
TP_F101U-00039: Tier1 Power Management User Guide 5G NAD
TP_F101U-00047:_User_Guide_Thermal_Management_5G_NAD

7.3 Website links

- Thermotron 2800 user manual: https://archive.org/stream/ThermotronIndustries-Thermotron-2800Programmer-Controller.-OperatorManual#page/n9/mode/2up
- Thermotron 3200 user manual: http://thermotron.com/pdf/_service/software-updates/3200-manual.pdf
- GPIB-USB controller download: http://www.ni.com/download/ni-visa-17.0/6646/en/
- GPIB-USB controller installation instructions: https://www.ni.com/getting-started/set-up-hardware/instrument-control/gpib-usb

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8 Revision History

Version	Date	Description
		Original document TP_F101U-00047_User_Guide_Thermal_Management_5G_NAD
01.00	Feb 22, 2023	Updated thermal zones indexes
02.00	Jun 26, 2024	Updated mitigation information.

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